Fintan Hardy

PhD Student, Imperial College London

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Summary

Third year PhD researcher in the Department of Materials at Imperial College London, working at the interface of experimental and theoretical materials science. My research focuses on accelerating microscopy studies of ferroelectric materials by developing advanced computer vision and deep learning models, and by integrating autonomous experimental workflows. I work across a range of AI methods including computer vision, generative models and causal inference, and I am keen to apply these approaches to challenging problems across the physical sciences and advanced technologies.

Education

- **Imperial College London** — **London, UK**
  PhD in AI for Materials Science (Supervised by Aron Walsh & Shelly Conroy) — *October 2024 – March 2027*
- **University Of Manchester** — **Manchester, UK**
  MEng (Hons) Chemical Engineering (**2:1**) — *September 2019 – July 2023*

Experience

Imperial College London — **London, UK**
PhD Student — *October 2024 – Present*

- Work focused on improvement of 4D-STEM dataset analysis using computer vision to uncover new phases and mechanisms in polarisation domain walls for next-generation nanoelectronics.
- Awarded the Student Researcher Award at Microscopy & Microanalysis 2025 (Salt Lake City), recognising outstanding early-career research.
- Have since released the preprint of my work: Polarization Domain Mapping From 4D-STEM Using Deep Learning

University of Manchester — **Manchester, UK**
Research Internship — *June 2022 – September 2022*

- Selected for a place on a competitive summer research internship supervised by Sam de Visser, using Density Functional Theory (DFT) methods to elucidate the reaction mechanism of deoxypodophyllotoxin, a promising anti-cancer drug component.
- First author on the resulting publication in Chemistry Europe (February 2024): Computational Study Into the Oxidative Ring-Closure Mechanism During the Biosynthesis of Deoxypodophyllotoxin.

iO Academy — **Bath, UK**
Software Developer Course — *July 2023 – October 2023*

- Achieved a government-funded position to take iO Academy's 12-week software developer course, learning how to write professional-level code, both individually and as part of a team. Projects included writing a fully functional front end and back end for a book store using JavaScript (React), PHP (Laravel) and Git.
- Learnt industry-standard coding practices such as SOLID principles for OOP, test-driven development and writing RESTful APIs, alongside regular practice presenting projects as part of an agile development environment.

Publications

2025: Polarization Domain Mapping From 4D-STEM Using Deep Learning (First Author, preprint)
2024: Computational Study Into the Oxidative Ring-Closure Mechanism During the Biosynthesis of Deoxypodophyllotoxin (First Author)

Conference Presentations

Oral Presentation: *Accelerated Identification of Polar Domains From 4D-STEM Using Group Convolutional Neural Networks.* Machine Learning and Generative AI for Microscopy session, Microscopy & Microanalysis 2025, Salt Lake City, USA.
Oral Presentation: *Accelerating Microscopy Workflows With Deep Learning.* Society of Electron Microscopy Technology Meeting 2025, Natural History Museum, London, UK.
Oral Presentation: *Ferroelectrics: From Latent to Lattice.* Future of Materials Design International Workshop (organised by the Walsh Group), National University of Singapore, 2026.
Oral Presentation: *Closing the Feedback Loop: Live Polarization Analysis for Automated In Situ 4D-STEM*. For the symposium: Living on the Edge: Real-Time Processing and Decision Making at the Microscope. Microscopy & Microanalysis 2026, Milwaukee, USA.
Oral Presentation: *Closing The Loop Towards Automated In Situ 4D-SEM and 4D-STEM Mapping*. For the symposium: Innovating Microscopy End-to-End: Hardware, Automation and AI in Honour of Albert Crewe. The 21st International Microscopy Congress, 2026, Liverpool, UK.
Upcoming: Oral Presentation at Materials Research Society Fall Conference 2026 (Boston, MA).

Teaching & Professional Development

Department Workshop, Imperial College London (2026): Designed and delivered a two-part lecture and tutorial on the fundamentals of machine learning and neural networks to postgraduate students and postdocs in the Department of Materials, co-delivered with Mathilde Franckel.
GeMSS Generative AI Summer School (2026): Selected and sponsored by AIchemy to attend intensive training on the fundamentals and advanced concepts behind generative models, alongside other early-career AI researchers.

Skills

Languages: Python (primary); C/C++ (built a small programming language compiler); prior experience with JavaScript (React) and PHP (Laravel)
ML & Deep Learning: *PyTorch*, computer vision, generative models, causal inference, deep learning
Data Science: NumPy, Pandas, SciPy, scikit-learn, Matplotlib.
Tools: *Git*, LaTeX, Linux, HPC / cluster computing